SAFETY DATA SHEET

According to EC 2020/878 (REACH) and 1272/2008 (CLP)

Last update: September 12, 2024

1. IDENTIFICATION OF THE SUBSTANCE/ MIXTURE AND OF THE COMPANY/ UNDERTAKING

1.1 PRODUCT IDENTIFIER

Product name: Filament Noctuo Nylon PA6

1.2 RELEVANT IDENTIFIED USES OF THE SUBSTANCE OR MIXTURE AND USES ADVISED AGAINST

Intended use: FDM 3D printing material

1.3 DETAILS OF THE SUPPLIER OF THE SAFETY DATA SHEET

Supplier:	**Manufacturer:**
Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska	Noctuo Sp. zo.o. ul. Sowińskiego 5 44-121 Gliwice, Polska

1.4 EMERGENCY TELEPHONE NUMBER

Emergency telephone: +48 42 63 14 724

2. HAZARDS IDENTIFICATION

2.1 CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

Classification according to Regulation (EC) No 1272/2008 of the European Parliament and of the Council (CLP):
This product is not classified as hazardous according to REGULATION (EC) No 1272/2008.

2.2 LABEL ELEMENTS

The product does not require labelling according to GHS criteria in accordance with Regulation (EC) No 1272/2008 of the European Parliament and of the Council.

Warning word: Not applicable

Hazard statement: There is no known significant impact.

Precautionary statement: Not applicable

2.3 OTHER HAZARDS

Persistent bioaccumulation and toxicity (PBT) or very persistent bioaccumulation (vPvB) is not possible because a chemical safety assessment is not required.

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 SUBSTANCE

3.2 PREPARATION

Nr CAS	Chemical name	Impurities Contributing to Hazard
25038-54-4	Polyamide 6	Nie sklasyfikowany

This product does not contain any additives or foreign substances required for product labelling and hazard classification according to REGULATION (EC) No 1272/2008 or contains regulatory amounts.

4. FIRST AID MEASURES

4.1 DESCRIPTION OF FIRST AID MEASURES

The product is stable under normal conditions, non-volatile. Warning before intervention; contact with hot product can cause severe burns. Inhalation of vapors may cause respiratory irritation and eye irritation may occur. In case of contact of hot material with eye or skin, appropriate medical intervention should be performed immediately.

Following eye contact:: Rinse carefully with running water. Remove contact lenses.

Following skin contact: In case of burns caused by melted product, immediately cool the burned area, preferably under running water. DO NOT remove the material from the skin. Removal may cause serious tissue damage. Seek medical attention

Following inhalation: Exposed to the inhalation of volatile compounds decomposition should be removed from the contaminated atmosphere, provide him with a supply of fresh air, protect from loss of heat; call a doctor.

Following ingestion: Seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

4.2 MOST IMPORTANT SYMPTOMS & EFFECTS BOTH ACUTE & DELAYED

There is no experience in humans that leads to acute or chronic damage..

4.3 INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED

Does not require special treatment.

5. FIRE - FIGHTING MEASURES

5.1 EXTINGUISHING MEDIA

Suitable extinguishing agents: Fire extinguishing foam, water spray, dry chemical, carbon dioxide.

Unsuitable extinguishing agents: Do not use water jets. Direct water jets on a burning product may cause a steam explosion and spread of fire. Avoid using foam and water on the same surface at the same time, as water destroys the foam.

5.2 SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE:

May be flammable at high temperature.
Combustion products: Carbon oxides (CO and CO2) and soot.
Combustion products may include thermo-oxidative decomposition products: carbon oxides, formaldehyde, acetaldehyde, organic acids (acetic acid), etc.
Combustion products may be toxic and/or irritating.

5.3 ADVICE FOR FIREFIGHTERS

Prevent re-ignition. If material is molten, do not contact water jet directly. Use water spray or foam. Special protective equipment for firefighters: Use self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

6.1 PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT & EMERGENCY PROCEDURES

Ensure adequate ventilation.

6.2 ENVIRONMENTAL PRECAUTIONS

Do not allow product to reach sewage system or water bodies. Uncontrolled discharges into watercourses must be immediately reported to the environmental protection authorities or other appropriate authorities. Waste dumping and its

waste should be collected in a timely manner and disposed of in specially designated places. Plastic waste is non-toxic and cannot be neutralized.

6.3 METHODS AND MATERIAL FOR CONTAINMENT AND CLEANING UP

Collect in suitable and appropriately labelled containers. Minimise dust generation during clean-up. Collect and place in suitable waste containers and seal tightly. Ensure adequate ventilation.

6.4 REFERENCE TO OTHER SECTIONS

For information on personal protection and waste disposal, see sections 8.2 and 13.

7. HANDLING AND STORAGE

7.1 PRECAUTIONS FOR SAFE HANDLING

Act in accordance with industrial hygiene and safety measures.
Avoid all sources of ignition.
Ensure grounding of handling equipment.
Do not swallow. Avoid contact with eyes.
Users should be protected from possible contact with molten product.
Do not eat, drink or smoke in the workplace.
Use personal protective equipment.
Processing equipment should be equipped with adequate ventilation.
Avoid inhaling fumes.
Do not exceed recommended operating temperatures. Do not incinerate.

7.2 CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES

Keep away from heat, sparks and flame. Protect from direct sunlight. Store in a dry, well-ventilated place.

7.3 SPECIFIC END USE(S)

The product is designed for use as a filament for 3D printing using FDM technology.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 CONTROL PARAMETERS

Traditional exposure control techniques can be used to effectively minimize worker exposure.

8.2 EXPOSURE CONTROL

Eye protection: Safety glasses with side shields.

Skin and body protection:	When working with molten filament, use additional gloves that protect against high temperatures.
Respiratory protection:	Respiratory protection in the presence of dust or insufficient ventilation. Combined filter for organic, inorganic, acidic inorganic and alkaline gases/vapours and toxic particles.
Hand protection:	Preventive skin protection.
Other hazards:	Users should be protected from the possibility of contact with molten material during processing.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Solid
Appearance:	Circular profile
Odour:	Not applicable
pH:	Not applicable
Melting point:	215-225 °C
Vapour pressure:	Not applicable
Vapour density:	Not applicable
Density:	1.14 g/cm3
Flash point:	350 °C
Decomposition temperature:	< 300 °C
Boiling temperature:	Not applicable
Solubility:	Insoluble in water
Solubility in solvents:	At room temperature, soluble: in concentrated acids, phenol, cresol.
Auto-ignition temperature:	<400 °C

9.2 OTHER SAFETY INFORMATION:

None.

10. STABILITY AND REACTIVITY

10.1 REACTIVITY

No hazardous reactions.

10.2 CHEMICAL STABILITY

Stable product if storage and handling requirements/instructions are followed.

10.3 POSSIBLE HAZARDOUS REACTION

None under conditions of normal use.

10.4 CONDITIONS TO AVOID

Temperatures above 300°C. Avoid prolonged melting and long-term storage at very high temperatures. To prevent thermal decomposition, filament processing should be carried out in the temperature range of 230 - 270 °C. The temperature of 290°C should not be exceeded.

10.5 INCOMPATIBLE MATERIALS

Acids and oxidizing agents.

10.6 HAZARDOUS DECOMPOSITION PRODUCTS

CO, HCN, NH3.

11. TOXICOLOGICAL INFORMATION

11.1 INFORMATION ON TOXICOLOGICAL EFFECTS

The product is non-toxic, has no side effects on human health.

Corrosive effect:	Not classified.
Mutagenic and carcinogenic effect:	The product is indifferent.
Sensitizing effect on the respiratory tract:	Inhalation of dust and/or thermal decomposition products may cause respiratory irritation.
Sensitizing effect on the skin:	Skin contact with molten/heated product may cause severe thermal burns.
Serious eye damage, irritation effect on the eyes:	The product may cause mechanical eyes irritation.

11.2 INFORMATION ON OTHER HAZARDS

No data.

12. ECOLOGICAL INFORMATION

12.1 TOXICITY

Under normal conditions, plastic is a very stable product. The product does not form toxic compounds with other substances in the air and water. The filament is poorly biodegradable, but does not pose a threat to the environment. It is practically insoluble in water and can therefore be separated from water.

12.2 PERSISTENCE AND DEGRADABILITY

No data available.

12.3 BIOACCUMULATIVE POTENTIAL

No data available.

12.4 MOBILITY IN SOIL

No data available.

12.5 RESULTS OF PBT AND VPVB ASSESSMENT

Persistent bioaccumulation and toxicity (PBT) or very persistent bioaccumulation (vPvB) is not possible because a chemical safety assessment is not required.

12.6 OTHER ADVERSE EFFECTS

No data available.

12.7 ADDITIONAL INFORMATION

Use the product in accordance with well-proven working methods, preventing its release into the environment.

13. DISPOSAL CONSIDERATION

13.1 WASTE TREATMENT METHODS

In accordance with local and national regulations.

14. TRANSPORT INFORMATION

Not classified as dangerous goods according to transport regulations (IMDG, IATA, ADR / RID).

14.1 UN number

Not applicable.

14.2 UN proper shipping name

Not applicable.

14.3 Transport hazard class(es)

Not applicable.

14.4 Packing Group

Not applicable.

14.5 Environmental hazards

Not applicable.

14.6 Special precautions for user

Not applicable.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable.

15. REGULATORY INFORMATION

15.1 SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS /LEGISLATION SPECIFIC FOR THE SUBSTANCE OR MIXTURE

EU Regulations; 2015/830/EC, 1272/2008/EC, 1907/2006/EC
Directives; 2006/8/EC, 2004/73/EC, 1999/45/EC, 67/548/EEC

15.2 CHEMICAL SAFETY ASSESSMENT

For this substance a chemical safety assessment is not yet required.

16. OTHER INFORMATION

The information contained in this sheet is based on current knowledge and experience with the product. This information is intended to support independent assessment of the methods used to ensure safe work with the filament and its safe disposal.